

QEM-5: The Upper Propagation Limit of Flip-Wave and the Causal Structure of Entanglement

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Abstract:

This work extends the Quantum Entanglement Mechanics (QEM) framework to explore the upper propagation limit of flip-waves within multi-body entanglement systems. We introduce a causal-cone model that defines the boundary of correlation spread and demonstrate that flip-wave coherence cannot propagate faster than the intrinsic coupling velocity of the phase lattice. This finding bridges the topology of entanglement, the structure of causal boundaries, and the limits of quantum-information transmission.

Series note: This work is part of the Quantum Entanglement Mechanics (QEM) series by Youfu Wang (2025).

1. Introduction

The QEM framework models entanglement as a mirror-coupled phase lattice that supports localized flips and collective flip-waves. Earlier parts established the layered mirror structure (QEM-8³) and stationary textures via domain walls (QEM-4). A central dynamical question is whether the propagation of flip-waves is bounded by an intrinsic speed, analogous to causal cones in relativistic field theories. Here we derive such an upper bound.

2. Upper Propagation Limit and Causal Cone

Let $\psi(x,t)$ denote the coherence amplitude between mirror-coupled nodes. We consider symmetric Σ and antisymmetric Δ modes. Linearizing around a stationary texture yields a wave-like equation for small flip perturbations with correlation length ξ and characteristic front velocity v_f .

Upper propagation limit:

$$v_p \leq v_{\max} = (J \xi)/\hbar.$$

Outside the cone $x > v_{\max} t$, the correlation $E(x,t)=|\psi|^2$ decays as $\exp[-(x-v_{\max} t)/\xi]$.

3. Physical Interpretation

Stability: The bound suppresses non-causal cross-talk.

Energy: $v_{\text{max}} \propto J \xi / \hbar$; finite-energy systems cannot exceed it.

Mirror reciprocity: Two-layer coupling enforces bounded Δ -mode propagation.

4. Implications and Experiments

Quantum communication: dynamic coherence remains causal.

Chip design: synchronize timing to $v_{\max}$; use domain-wall textures for protected storage.

Experiments: (i) extract ξ from tanh wall; (ii) standing texture with sech^2 envelope; (iii) track front to estimate $v_{\max}$.

5. Conclusion

QEM-5 establishes an intrinsic upper propagation limit for flip-waves and formalizes the causal cone. It connects structure (QEM-8³), stability (QEM-4), and dynamics (this work), enabling prototype design (QEM-6).

References

1) Wang, Y. (2025). QEM-4: Topological Domain-Wall and Stationary Interference Texture in Multi-Body Entanglement. Zenodo. DOI: 10.5281/zenodo.17540435

2) Wang, Y. (2025). QEM-8^3: Hierarchical Mirror Lattice in Multi-Body Entanglement. Zenodo. DOI: 10.5281/zenodo.17536955